



Bit Manipulation Problem Solving - 3

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Problem 1

Maximum AND:

<https://www.codechef.com/problems/MAXAND18>



N
 $A_1 \quad A_2 \quad \dots \quad A_n$

$X \rightarrow \underline{K \text{ set bits}}$

$$S = (A_1 \oplus X) + (A_2 \oplus X) + \dots + (A_n \oplus X)$$

Aim $\rightarrow$ maximize value of S . $x_1 \ x_2 \ x_3$

If multiple $X \rightarrow$ choose the
smallest value of X



$X \rightarrow$ 0 0 0 1 0 1 0 1 0

$X \leq 10^9$

31 bits

K bits

$K = 3$

~~maximize S~~ ✓
↓
minimize X ✓

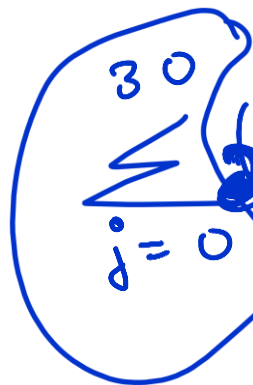
$x \rightarrow \underline{\underline{31}}$

$S = \ominus \ominus \ominus \ominus \ominus \ominus$



$\textcircled{S \neq}$

$$\sum_{i=0}^{n-1} (A_i \& x)$$



(jth bit in x)

$\times$

$$\sum_{i=0}^{n-1} (j^{\text{th}} \text{ bit in } A_i)$$



$$2^j \cdot x$$



* 5 numbers that have bit 0 as a set bit



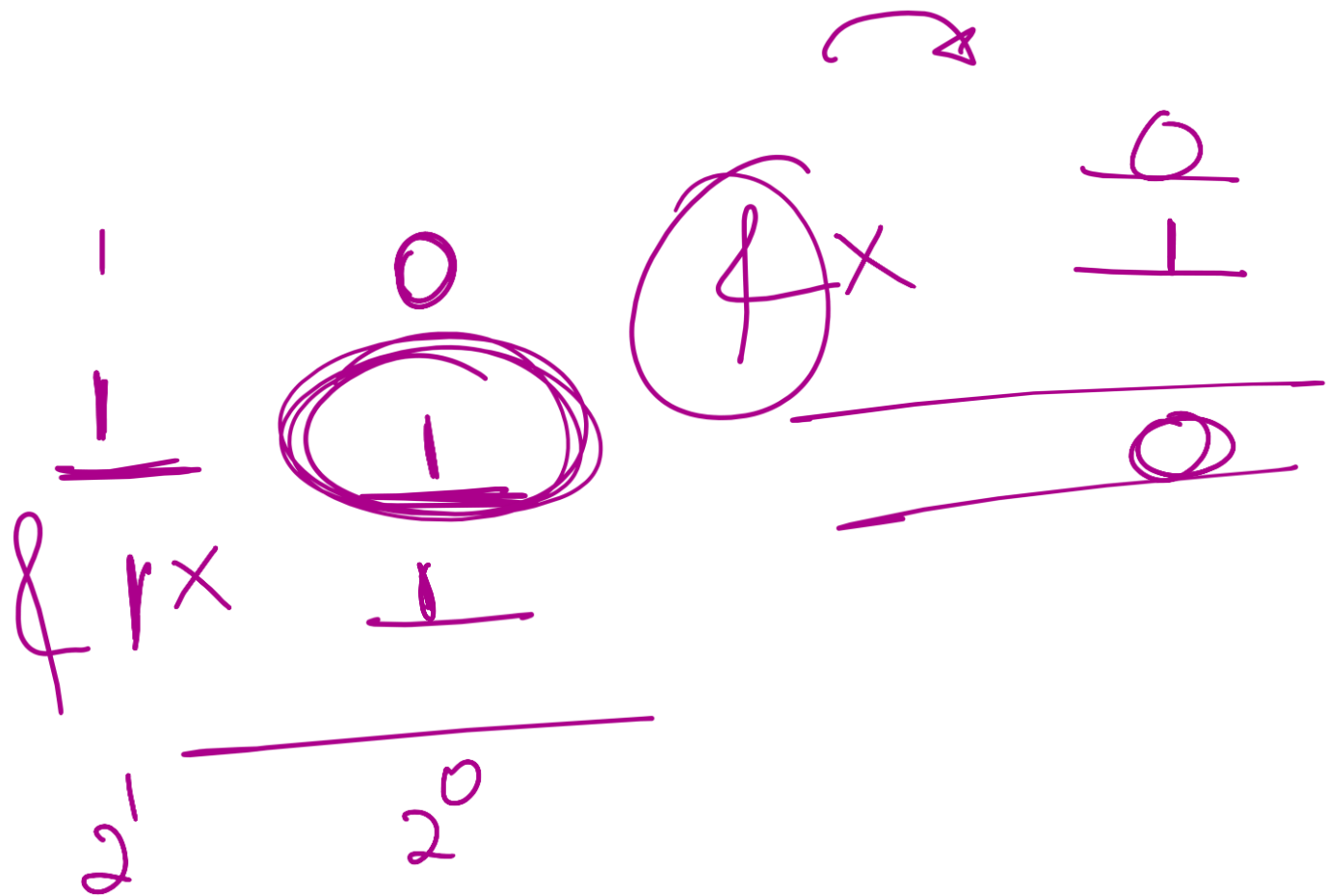
Contribution from bit 0 if X has a

$$\text{set bit} = (2^0 \times S) \quad (2^y) \times (\text{rectangle})$$

* Contribution from bit 0 if X does not have a set bit at position 0 = 0

$$X \& A_i$$

$$S = \sum X \& A_i$$



$$\underline{\underline{A_i \leq 10^9}}$$



A_1

— — — — — . . . — — —

A_2

— — — — — . . . — — —

A_3

— — — — — . . . — — —

A_n

— — — — — . . . — — —

$\times \& A_i$

Try to think bit by bit.

$x \rightarrow 0$ set



(n)

5 numbers that
have bit 0 set bit



$$\boxed{2^0 \times 5}$$

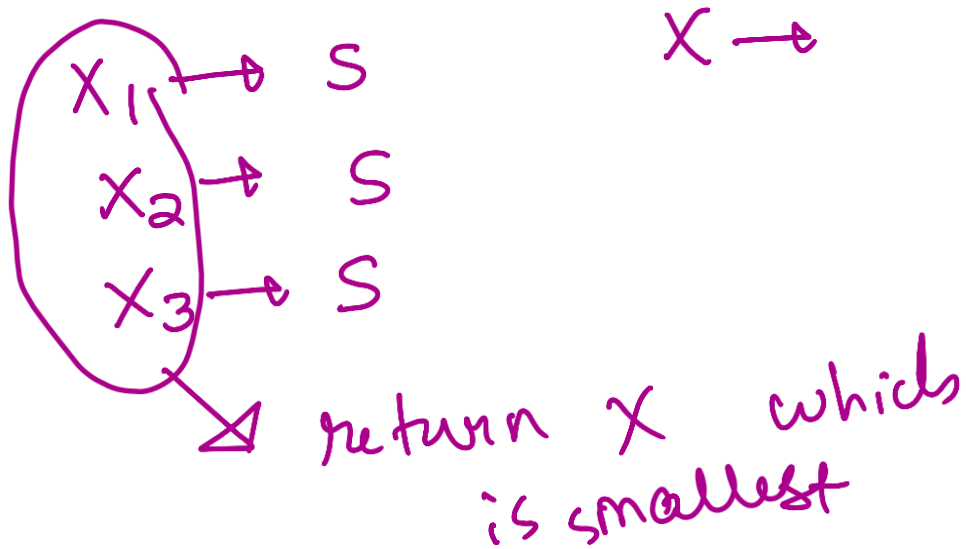
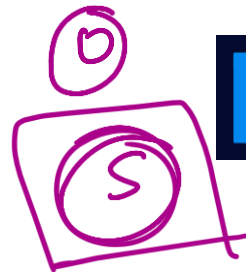
3 numbers
that have bit 1 set
bit



$$\boxed{2^1 \times 3}$$

K bits that can be set in x.

31
0, 1, 2, - - - - 30



2 set bits

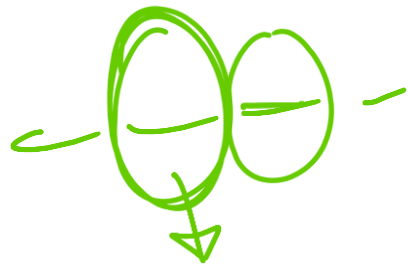
$$k=2$$

X



X min

bit 0 → 5
✓ bit 1 → 10 ✓
✗ bit 2 → 10 ✓
bit 3 → 15 ✓
bit 4 → 30 → 0





Sorting criteria

→
Sort in descending order according to contribution.

∴
For same contribution
sort in ascending order
of bit position



$(15, -3), (10, -2), (10, -1), (5, 0)$

$(15, 3), (10, 2), (10, 1), (5, 0)$

$(15, -3), (10, -1), (10, -2), (5, 0)$

$(15, 3), (10, 1), (10, 2), (5, 0)$

$(15, 3), (10, 2), (10, 1), (5, 0)$

X



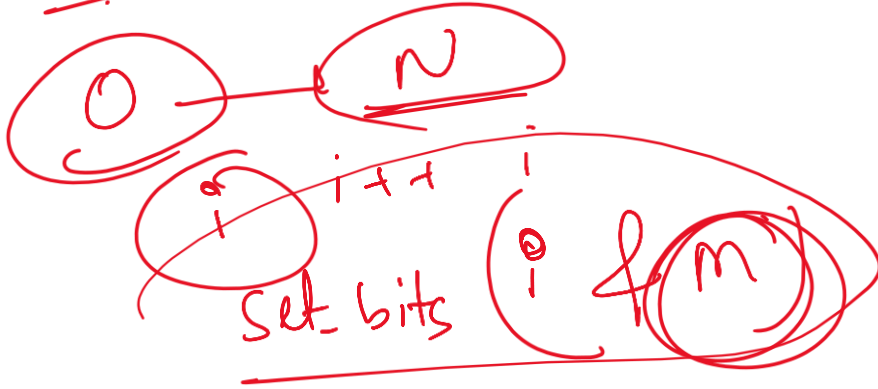
Problem 2

Masked Popcount:

https://atcoder.jp/contests/abc356/tasks/abc356_d



N, m



Bit by bit



$\sum_{i=0}^N$ set bits($i \& m$)



tell me what is the
contribution from this bit
in the summation

if bit is set in $m \rightarrow$ # of numbers from $0-N$ that
have i^{th} bit as set

if bit is unset in $m \rightarrow 0$

For Any bit i , tell
me the # of numbers $0 \rightarrow n$
that have i^{th} bit as a
set bit



$$0 \leq n \leq 2^{60}$$

$M \rightarrow$

i

2^i

K

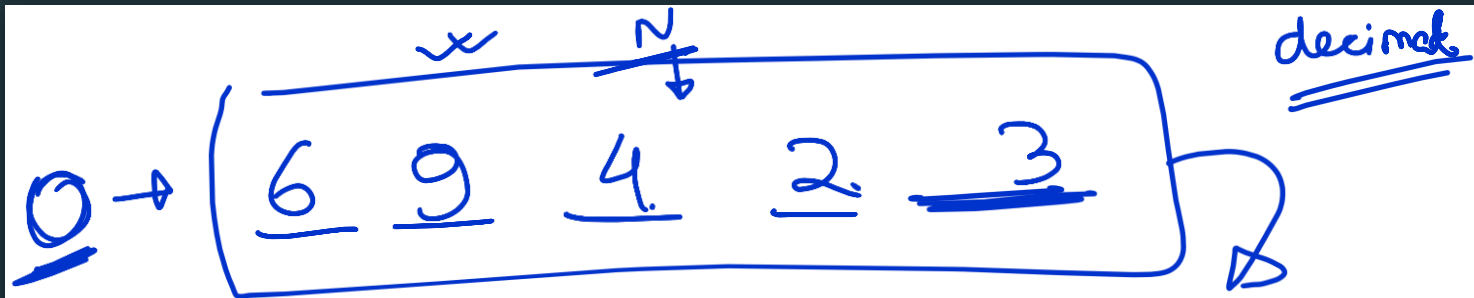
i

i

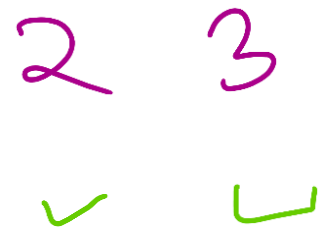
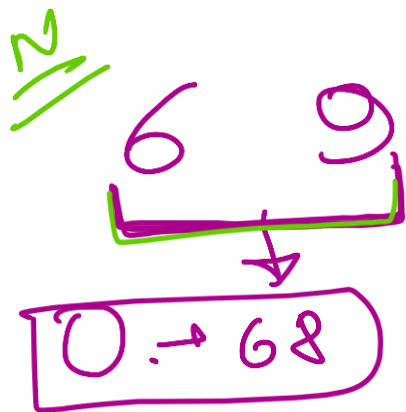
K^i

$\frac{0}{0}$

$1 + 0 = 1$



— — 8 — —



2



8



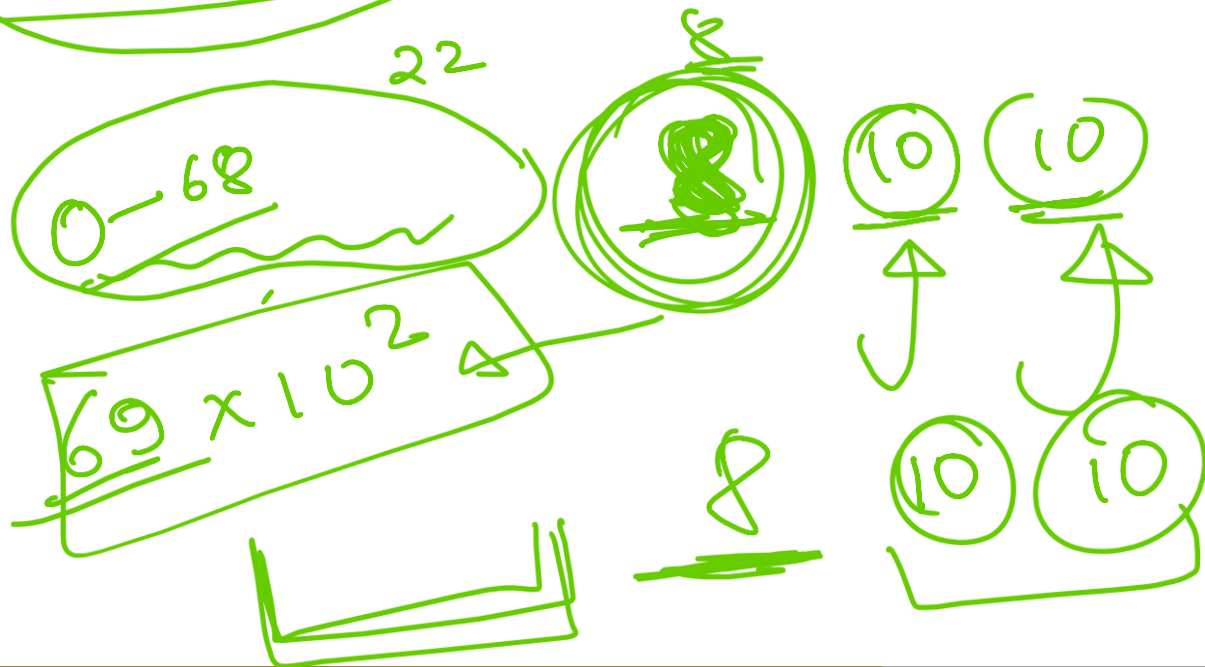
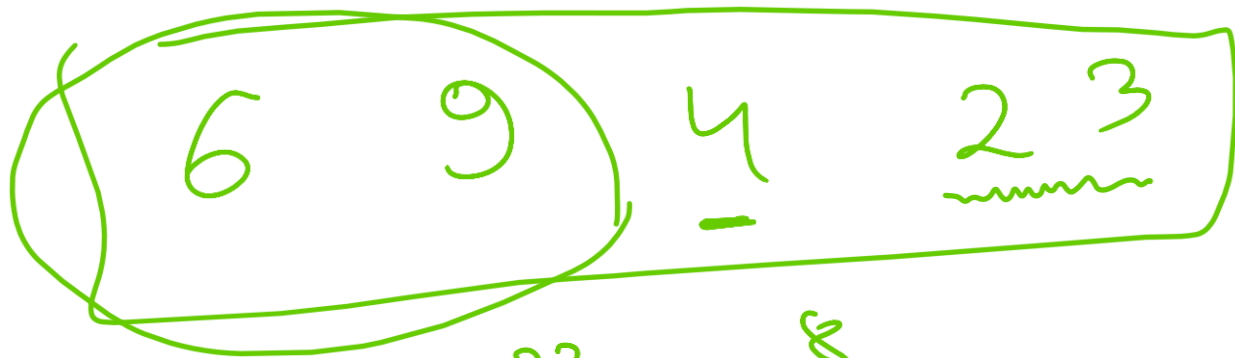
8 6 2 4

859

0-7 5 0-9

7 5 9 →
→
0-7

≤ 2



694.23

69

0.68

8

70

8x10

0-7

8

~~8~~

6

5

8 5 0
8 5 1
8 5 2
...
8 6 2

2

10

$\leq N$

$N \rightarrow 858$

~~8~~ 5 (9)

0 $\rightarrow$ 8

850 → 858

0 → 2



0 → 7

8

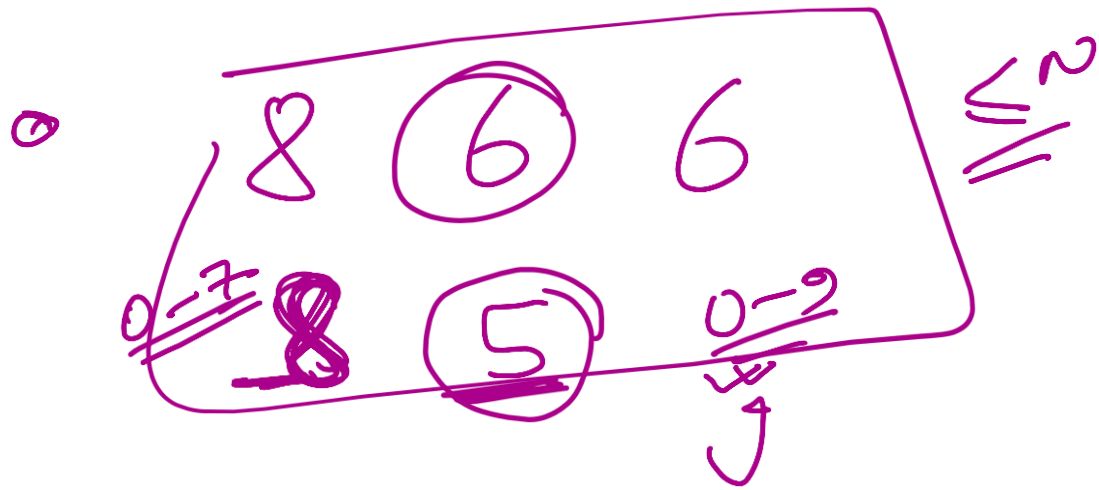
5

0/1/2... 8



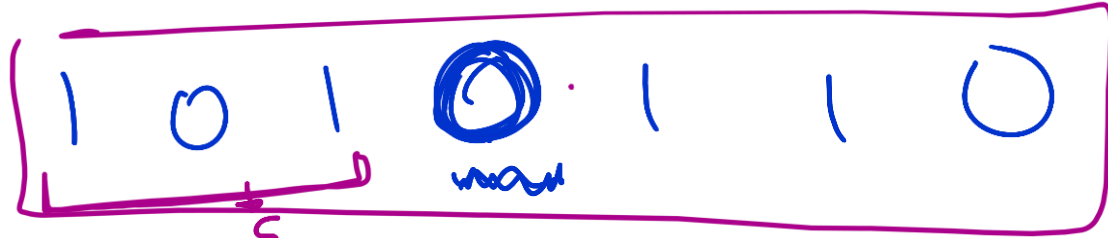
$$\begin{array}{r}
 \boxed{8 \quad 4 \quad 8} \quad 2 \\
 \hline
 0-7 \quad 8 \quad 5 \quad 9 \times
 \end{array}$$

$$\begin{array}{r}
 8 \times 10 \\
 \hline
 80
 \end{array}
 +$$



$$(8 \times 10) + 10 = 90$$

2/21



2/21



[0-4]

1

2

2

2

$5 \times 8 = 40$ → 40



$$N = \boxed{10101} \quad \underbrace{110}_{0-6} \quad \underline{0/1/2 \dots 6}$$

10 6

2 6

$$\underline{\underline{0-9}}$$

$$\boxed{10 \times 8 = 80}$$

$$10$$

$$80 + \boxed{7} = \underline{\underline{87}}$$

$$1$$

$$\underline{\underline{7}}$$

0-6

1 0 1 0 1 0 0 0

1 0 1 0 1 0 0 1

1 0 1 0 1 0 1 0

1 0 1 0 1 0 1 1

1 0 1 0 1 1 0 0

1 0 1 0 1 1 0 1

1 0 1 0 1 1 1 0

1 0 1 0 1 1 1 1

7

$$N = \underbrace{10}_{\downarrow} \begin{matrix} \textcircled{S} \\ \text{---} \\ \text{---} \end{matrix}$$

$$\boxed{S \times 2^2}$$

$\textcircled{i} \rightarrow 0\text{-based format}$

1 0 0

$$\boxed{(N \gg (i+1)) \ll i}$$

$N \rightarrow 101100$

$N \gg (i+1)$

000101

$\ll i$

010100

40



1 = 2

6-4

$$(i+1)$$

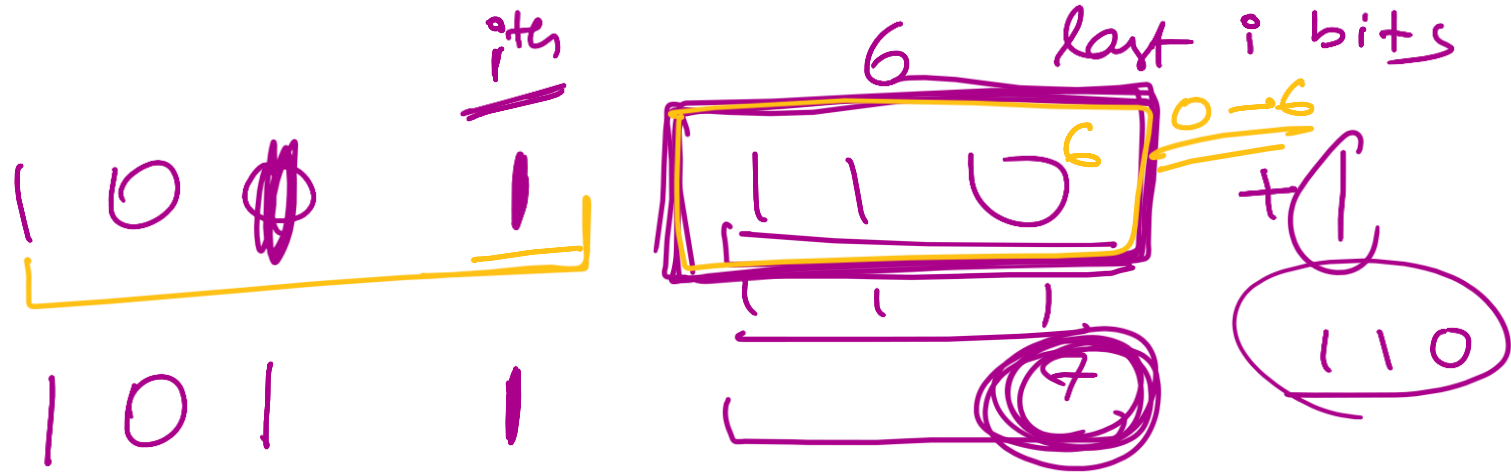
$(S \times 2^2) \times 4$

5×2^2

$S \times 2^0$

$$N \gg (i+1)$$

LOI



way 1 $\rightarrow N \& (2^i - 1)$ ✓

way 2 $\rightarrow N \% 2^i$